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CLINICAL RESEARCH

Comparison of primary and secondary enucleation for uveal melanoma

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ABSTRACT

We investigated operative course and post-operative findings of patients undergoing primary enucleation for uveal melanoma versus those requiring secondary enucleation after brachytherapy. A retrospective chart review was performed with IRB approval on patients receiving treatment for uveal melanoma. Patients with enucleation as initial treatment and patients enucleated after plaque brachytherapy were analyzed for demographic data, operative course, and post-enucleation outcome. Further cause analysis for secondary enucleations was investigated. No significant difference was seen in age, laterality, or gender between the primarily enucleated (n = 54) and secondarily enucleated (n = 34) groups. Greater difficulty with surgery was noted in 28/32 (87.5%) of secondary enucleations compared to 1/54 (1.8%) of primary enucleations ($p < 0.0001$). Operative time was >2 hours in 3/51 (6%) of primary enucleations (vs. 8 of 32, 25%, $p = 0.02$). Average implant size was similar in the 2 groups (20.6 mm), however 2/34 (6%) of secondary enucleations required dermis fat grafting. Post-enucleation anophthalmic ptosis occurred after 8/49 (16%) of primary cases (vs. 13/30, 43%, $p = 0.02$) and prosthetic enophthalmos after none (0%) of primary cases (vs. 5/30, 17%, $p = 0.006$). Class 2 gene expression profile was found in 6/8 (60%) of eyes enucleated for treatment failure. Secondary enucleation performed after plaque brachytherapy was technically more difficult, and had more anophthalmic socket and eyelid complications compared to primary enucleation for uveal melanoma. Primary enucleation may avoid additional surgery and morbidity in a subset of patients with contraindications to plaque brachytherapy.

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Introduction

Uveal melanoma is the most common adult intraocular malignancy with a stable incidence of about 5.1 per million per year in Caucasians.[1] There are several treatment modalities including primary enucleation and globe-conserving therapies such as plaque radiotherapy with iodine-125 (I-125) or ruthenium-106, stereotactic proton or photon beam therapy, or local resection.[2-10] The Collaborative Ocular Melanoma Study (COMS) established that mortality from medium-sized choroidal melanomas after treatment with plaque radiotherapy with I-125 did not differ significantly from primary enucleation.[2,3]

Since that time brachytherapy with I-125 has become increasingly popular as a treatment modality particularly for its potential to decrease visual morbidity and allow globe preservation. Radiotherapy is not without its complications however, and I-125 brachytherapy is associated with complications such as keratitis, cataract, iris neovascularization, neovascular glaucoma, radiation retinopathy, and optic neuropathy.[11]

Sagoo et al. found that 66% of post-plaque patients with posterior uveal melanoma suffered from radiation retinopathy, 66% had cataract, and 20% required enucleation by 5 years after plaque.[12] Although there is literature examining visual morbidity and overall survival in patients undergoing plaque brachytherapy for uveal melanoma, to our knowledge there are none that examine operative course and anopthlamic complications in the subset of patients who require enucleation after plaque. The current study was undertaken to investigate the operative course and post-operative findings in patients undergoing primary enucleation versus those requiring enucleation after I-125 plaque radiotherapy.

Methods

A retrospective chart review was performed under IRB approval at Washington University in Saint Louis, Saint Louis, Missouri, USA (#201312030) on patients receiving treatment for uveal melanoma between 1992–2014. Patients that underwent enucleation as initial therapy as

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well as those that needed enucleation secondarily after plaque brachytherapy with I-125 were identified. Both groups were analyzed for demographic and clinical data, operative course, and post-enucleation outcome. Further cause analysis for secondary enucleations was investigated.

Treatment failure was defined as evidence of tumor non-response, recurrence, growth, or evidence of extra-scleral extension not present prior to plaque placement. All gene expression profile (GEP) data was obtained via trans-scleral or trans-vitreal biopsy at the time of plaque placement. Enucleations were performed in the standard technique with attachment of all extraocular muscles to a donor sclera covered acrylic or porous polyethelyene implant, followed by layered Tenons and conjunctival closure. In the patients requiring dermal fat grafting, muscles were attached to the periphery of the dermis with conjunctiva attachment along the grafts anterior edge. Anophthalmic ptosis was defined as a margin-reflex distance (MRD1) difference ≥ 2 mm with prosthesis in place, and enophthalmos was defined as ≥ 2 mm difference in Hertel measurement with prosthesis in place. Statistics were performed with Fisher's exact test for discrete data, and Mann–Whitney for continuous data.

Results

Of the 416 patients identified, 2 were excluded for non-Caucasian race, 15 were excluded for age <30 years or >85 years, 61 pursued alternative therapies or were lost to follow-up prior to treatment, 54 patients had enucleation as initial treatment, and 284 underwent plaque brachytherapy with I-125. Follow-up from time of plaque placement to enucleation, last follow-up, or end of study ranged from 1 month to 20 years and averaged 56.8 months (median 41 months, SD 43 months). Average time to enucleation was 4.7 years (range 1–20 years, SD 4.2 years) with the majority of eyes not requiring enucleation by 10 years (Figure 1).

Thirty-one of the plaque patients eventually required enucleation and an additional 3 were included that underwent plaque at an outside institution. No statistically significant difference was seen in age at enucleation, laterality, gender, or gene expression profile (GEP) between the primary and secondary enucleation groups (Table 1). Tumors in the primary enucleation group were more likely to be large size (38/46, 83% vs 9/29, 33%; $p < 0.0001$) with greater initial tumor thickness (9.6 mm vs 5.5 mm, $p < 0.0001$). Radiation dosage was available for 14/34 patients. Average scleral dose was 329.5 Gy (range 186 Gy-505 Gy, SD = 115.63). The 2 cases where dermis fat grafting was required had scleral dose of ≥475 Gy.

Average operative time for primary enucleation (83 minutes, range 51-126) tended to be shorter than secondary procedures (100 minutes, range 54–177) ($p = 0.99$). Operative time was over 2 hours in 3/51 (6%) of the primary enucleations vs 8/32 (25%) of those performed secondarily ($p = 0.02$). Operative reports notating scarring or fibrosis requiring extended or sharp dissection were found in 28/32 (87.5%) of secondary enucleations compared to 1/54 (1.8%) of primary enucleations ($p < 0.0001$). Average implant size was similar

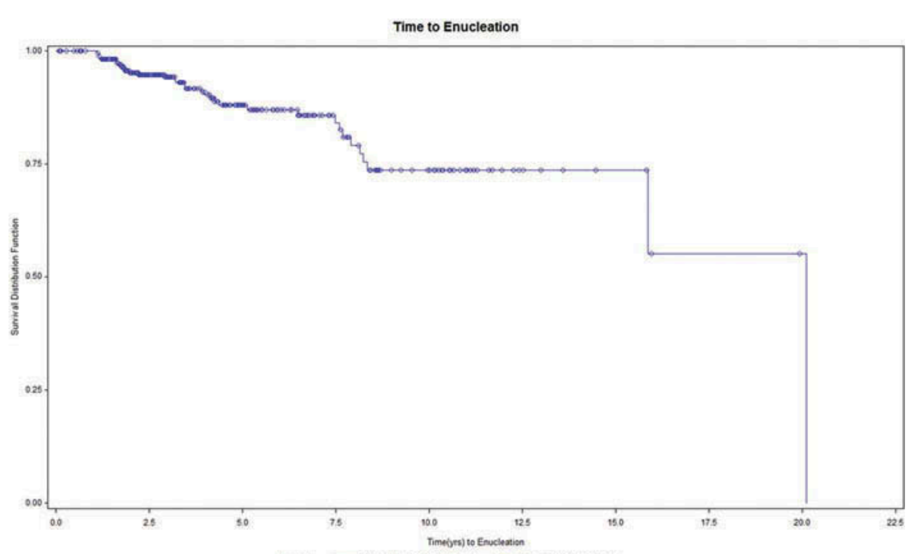

Figure 1. Kaplan–Meier survival curve of time from placement of I-125 plaque to enucleation.

Table 1. Demographic and clinical characteristics.

		Primary (n)	Secondary (n)	P-value
Average age at enucleation		61 (54)	66 (34)	0.23
Average months follow-up		24 (49)	17 (30)	0.79
Gender	M	50% (27)	62% (21)	0.38
	F	50% (27)	38% (13)	
Laterality	OD	44% (24)	47% (16)	0.83
	OS	56% (30)	53% (18)	
GEP	1	24% (9)	37.5% (6)	0.34
	1B or 2	76% (28)	62.5% (10)	

M – male; F – female; OD – right eye; OS – left eye; GEP – gene expression profile.

in the two groups (20.6 mm) however 2/33 (6%) of secondary enucleations required dermis fat grafting (Figure 2).

Post-enucleation anophthalmic ptosis was noted in 13/30 (43%) of the secondary group (vs 8/49, 16%; p = 0.02). Prosthetic enophthalmos (5/30, 17% vs 0/49, 0%; p = 0.006) and socket volume deficit (3/30, 10% vs 0/49, 0%; p = 0.05) were also more common in the secondary group. Additional procedures such as correction of eyelid malposition and removal of conjunctival cyst were more commonly performed in secondary patients (3/30, 10%) than those undergoing primary enucleation (3/49, 6.12%) (p = 0.67) (Table 2). There was only 1 case of implant extrusion, which occurred in the primary enucleation group.

In the group requiring enucleation after plaque, 18/34 (53%) were for treatment failure while the others were enucleated for multifactorial reasons - primarily pain and/or complications from neovascularization (15/16, 94% and 14/16, 88%, respectively). This group had more males than females; however, this was not statistically significant (21/34, 62% vs 13/43, 38%, p = 0.28). Average time to enucleation was shorter in the treatment failure group (4.5 vs 6 years, p =0.83) compared to those enucleated for other reasons. Overall 14/34 (40%) of the secondary enucleations occurred in the first 3 years after plaque and a higher proportion of enucleations in the treatment failure group were performed during this time frame (8/18, 44% vs 6/16, 38%; p = 0.74). In those where GEP data was available, a higher proportion of patients enucleated for treatment failure had class 2 GEP compared to those enucleated for other reasons (6/10, 60% vs 0/6, 0%; p = 0.03).

Discussion

Even though orbital recurrence after primary enucleation is rare and appears in the literature on the order of case reports while recurrence after plaque necessitating either further treatment or enucleation has been reported to be ~10%, globe-conserving therapies have become increasing popular.[13-16] The appeal of plaque brachytherapy in the treatment of uveal melanoma is the preservation of globe and vision. It has been found that 34% of patients retain visual acuity (VA) better than 20/40 at 3 years after plaque brachytherapy, however VA declined at a rate of ~2 lines per year and by 10 years almost 90% of patients with juxtapapillary tumors had vision worse than 20/200.[12,17]

In addition to an accelerated decline in visual acuity, complications such as radiation retinopathy and neovascular glaucoma often result in a blind painful eye and enucleation is eventually required in some cases. In the current study population, enucleations after plaque were required in 31/301 (10.3%) of eyes with other authors finding the rate to be 6–12.5%, with one study reporting a rate of almost 20% for large posterior melanomas.[12,18-21] Post-plaque enucleations were

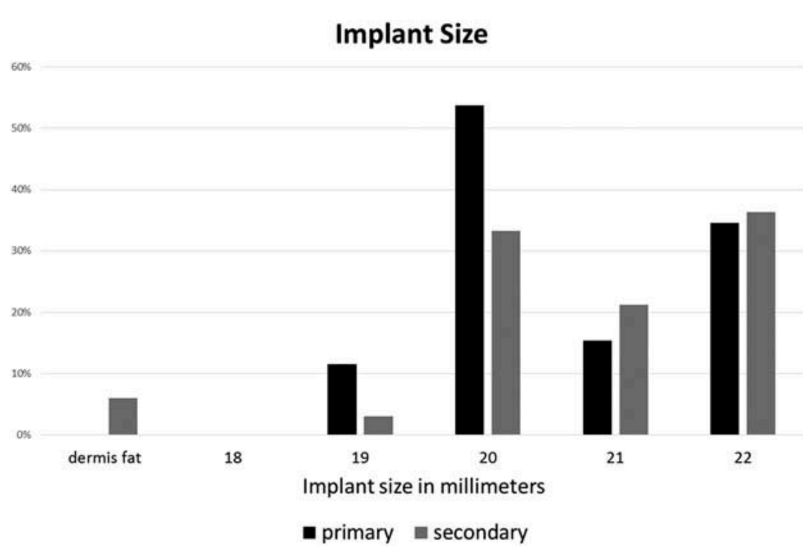

Figure 2. Implant size.

Table 2. Additional procedures and/or surgeries between plaque and enucleation.

	Primary enucleation (N)	Enucleation after plaque (N)	P-value
Anophthamic ptosis	16.3% (8)	43.3% (13)	0.02
Entropion/ectropion	2% (1)	6.7% (2)	0.55
Enophthalmos	0	16.7% (5)	0.006
Volume deficit	0	10% (3)	0.05
Fornix foreshortening	0	6.7% (2)	0.14
Total number of patients	(49)	(30)	

performed for treatment failure in 18/34 (53%) of the current group, similar to published reports of 35–61%.[2,16-21]

The endpoint of enucleation is reached by as many as 1/5 of eyes undergoing plaque, and the current study finds that enucleations performed secondarily are more difficult, time consuming, and have more post-enucleation socket and eyelid deformities when compared to eyes undergoing enucleation primarily.

In the present group, operative reports of secondary enucleations were more likely to make notation of a particularly difficult surgery due to scarring and fibrosis necessitating sharp dissection (28/32; 87% vs. 1/54; 2%). Surgeons felt that intraoperative scarring was excessive in irradiated eyes compared with enucleations for eyes that have undergone extensive surgery. There was a trend towards longer enucleation operative times required with increase in scleral radiation dose, however given the small sample size with high standard deviation and low coefficient of determination on simple linear regression analysis, further study with larger sample sizes would be needed to better delineate this relationship (Figure 3).

It was hypothesized that increased scarring and fibrosis would lead to conjunctival shortage requiring smaller implant size in the secondary group; however, it was found that average implant size was the same in both (20.6 mm). There may be several reasons for this finding. Post-plaque orbital tissues may have experienced atrophy thus increasing the size of implant needed to maintain socket integrity. Additionally, 2/34 (6%) of secondary enucleations required primary DFG that were not included in implant size analysis. Not only does DFG result in an additional site of post-operative pain and nidus for surgical site infection, but in the current study resulted in longer operative time (operative times of enucleations requiring DFG were over 2 h with one lasting nearly 3 h).

In the United States, 87% of enucleations are performed under general anesthesia (GA) and the anesthesia literature has shown a significant correlation between duration of GA and post-operative morbidity and mortality.[22-26] In the current study operative times were found to be 17 minutes longer in the secondary enucleation group (100 min vs 83 min, p = 0.99). The authors suspect that the ability to quantify the primarily physician-in-training level of expertise involvement in the primary enucleations (which presumably falsely elevated average operative time in the uncomplicated enucleation group) as compared to the post-graduate attending level expertise required for the secondary enucleations would have yielded a more notable difference. Despite no difference in average time, there were a significantly larger proportion of cases requiring over 2 hours in the secondary group (8/32; 25% vs 3/51; 6%, p = 0.02). It can be hypothesized that this would increase risk of post-operative complications, however whether this correlates to a significant increase in morbidity would require further analysis.

In contrast to a retrospective series by Fabian et al., where there were no significant socket complications or issues with comesis,[27] a higher rate of anophthalmic socket and eyelid deformities such as enophthalmos

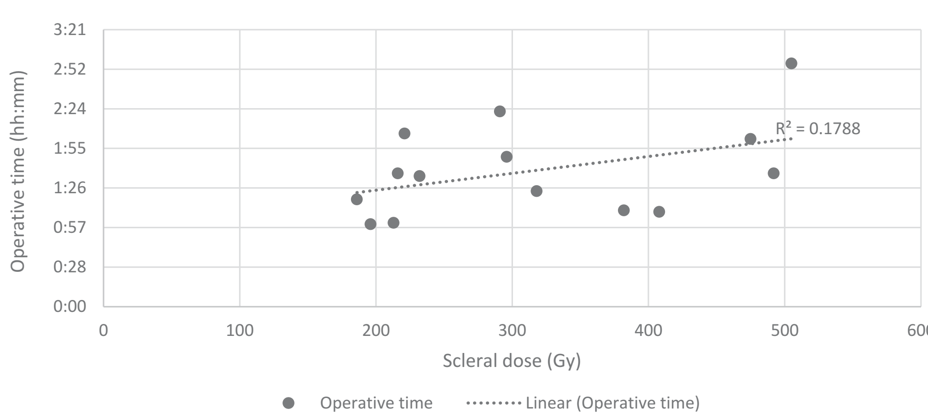

Figure 3. Relationship between scleral dose and enucleation operative time (n = 14). Operative time ranged from 1:00 to 2:57 (hh:mm), SD 0:33. Scleral dose ranged from 186 Gy to 505 Gy, SD 115.63 Gy.

and ptosis was observed in our series. The difference in these findings may reflect differences in primary treatment as the Fabian series included patients who underwent Ru-106 plaque brachytherapy, proton beam therapy, and transpupillary thermotherapy, and quantification of eyelid position by MRD in that series may have yielded similar results to our series.

Factors such as post-plaque radiation effect as well as the need for DFG may play a role in socket and eyelid deformities. Both cases requiring DFG received scleral doses ≥475 Gy, the highest and third highest radiation doses in our series. Smith et al. observed 25% volume atrophy of DFG while Yu et al. found 27% reduction in volume and clinical enophthalmos by 9 months (presented at the Hong Kong Ophthalmological Society Annual Scientific Meeting on December 9, 2010).[28]

Even though the secondary group had a significantly higher rate of socket and eyelid deformity after enucleation, the number of additional corrective surgeries was not statistically different between the two groups. This in part may be due to "surgical fatigue" such that patients who have already undergone multiple procedures may be less likely to opt for further surgery. In our series, 18 of the 34 (53%) post-plaque patients underwent some sort of procedure or surgery between plaque and enucleation (cataract and pars plana vitrectomy were the two most common with 10 and 11 occurrences, respectively) with a median of 1 additional procedure and two patients undergoing as many as 5 additional procedures (Table 3).

Cytogenetic studies have greatly improved prognostication in uveal melanoma and it has been shown that class 2 GEP strongly correlates with metastasis and decreased survival.[29-31] As of 2012 82% of ocular oncologists performed some sort of cellular and/or molecular analysis and 94% were performed at the time of plaque placement or enucleation.[32] Although GEP status changed post-treatment management for 79% of survey respondents, this information is not currently being used to guide initial treatment decisions.[32] We found that post-plaque brachytherapy eyes enucleated for treatment failure were more likely to have class 2 GEP. Knowledge of GEP status prior to initial treatment decisions may help identify individuals at higher risk of post-plaque treatment failure and early secondary enucleation.

Despite early preservation of form and function using globe-conserving therapies in the treatment of uveal melanoma, a number of patients eventually undergo enucleation. The increased duration of time under general anesthesia associated with these secondary enucleations may place the patient at higher risk of postoperative morbidity and mortality. The higher likelihood dermis fat grafting in secondary enucleations likely contributes to the higher proportion of anophthalmic socket and eyelid complications. Similar to the COMS finding of treatment failure being the number one reason for enucleation in the first 3 years,[19] a subset of patients has been observed to require enucleation within 3 years after plaque —these patients were more likely to be enucleated for treatment failure and to have class 2 GEP. The ability to obtain GEP data prior to initial management decisions may help identify those at higher risk for treatment failure, early enucleation, and anophthalmic socket and eyelid complications.

Table 3. Procedures performed between plaque placement and enucleation.

Procedure	Number of occurrences	Number of patients
Cataract extraction	10*	10
Pars plana vitrectomy (all reasons)	11	8
Scleral buckle	1**	1
Anterior chamber washout	3	2
Diode cyclophotocoagulation	3	2
Laser retinopexy	2	1
Scleral patch graft	2	1
Strabismus	2	2
Argon laser for failure	3	1

* One was a complicated cataract surgery.
** Placement of scleral buckle was attempted, but aborted due to excessive scarring.

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Disclosure statement

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the article.

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